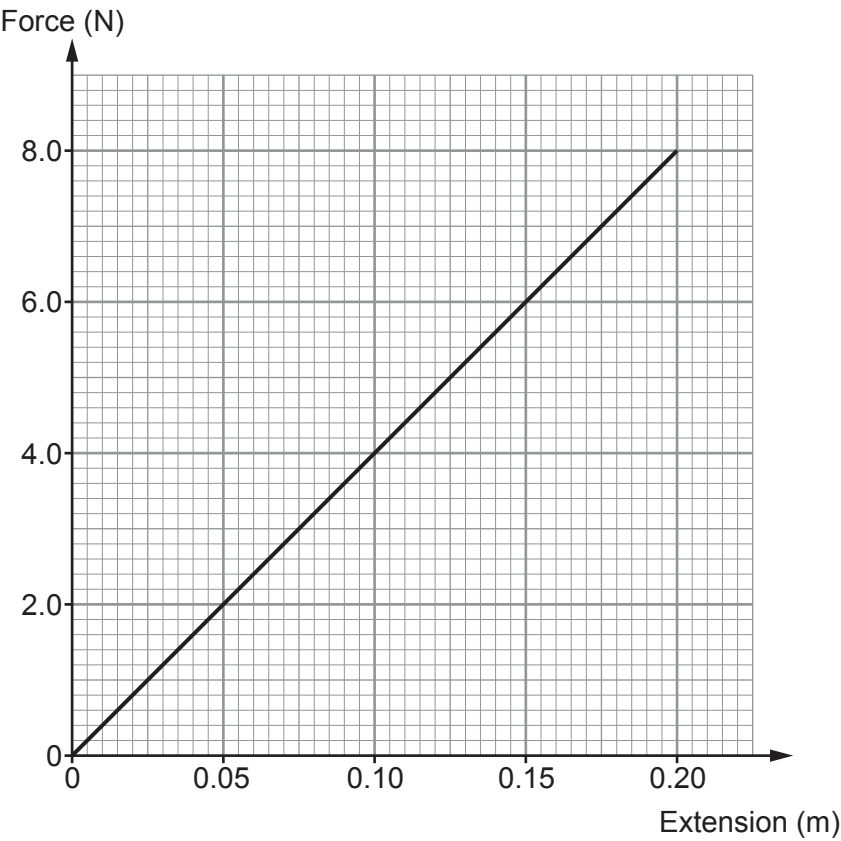


3. (a) Students carry out an experiment on a spring. The length of the spring is measured for each force and extensions are calculated. The force-extension graph is shown below.



Use the equation:

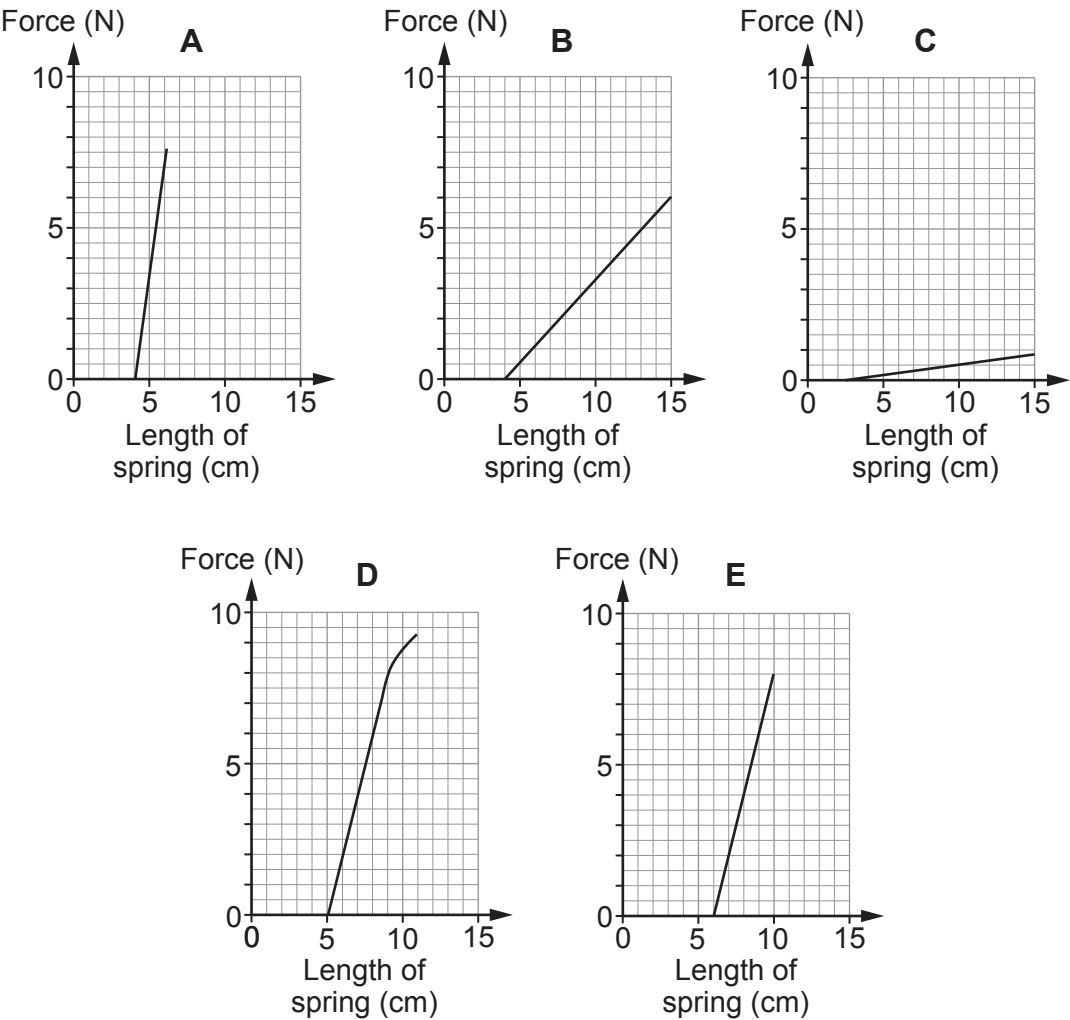
$$\text{spring constant} = \frac{\text{force}}{\text{extension}}$$

and the graph to calculate the spring constant in N/m for a force of 8.0 N. [2]

spring constant = N/m



- (b) The following graphs were obtained when students in a class used forces to stretch five different springs, **A**, **B**, **C**, **D** and **E**. These students measured the lengths of each spring with different forces and did not calculate extension. All graphs are drawn on axes with the same scales.

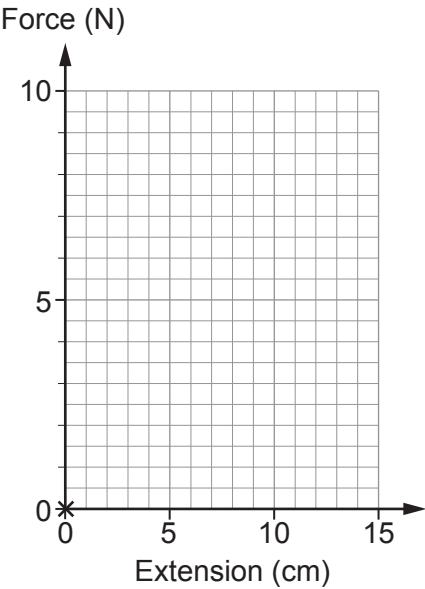


- (i) Use the graphs above to answer the following questions about springs **A** to **E**.
- I. State which spring is easiest to stretch. [1]
 - II. State which spring has the biggest unstretched length. [1]
 - III. State which spring has the same spring constant as **D**. [1]
 - IV. State which spring is stretched beyond its elastic limit. [1]



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only

- (ii) They decided to calculate extensions and draw the **force-extension** graph for Spring **E**.
Draw this graph on the axes below.
The first point has been plotted for you. [2]



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